AI-Powered Data Cleansing & Diagnostics Report

Goal: i dont want date and quantity column remove it

1. Executive Summary

This diagnostics report details the cleaning operations performed on dataset EV_Dataset.csv. Guided by user goals and Nvidia GLM-5.2 recommendations, duplicate/redundant structures were dropped, datatypes were normalized, and missing values imputed.

Dimension	Original Dataset	Cleaned Dataset
Total Rows	96845	96845
Total Features / Columns	8	6

2. Applied Feature Rules

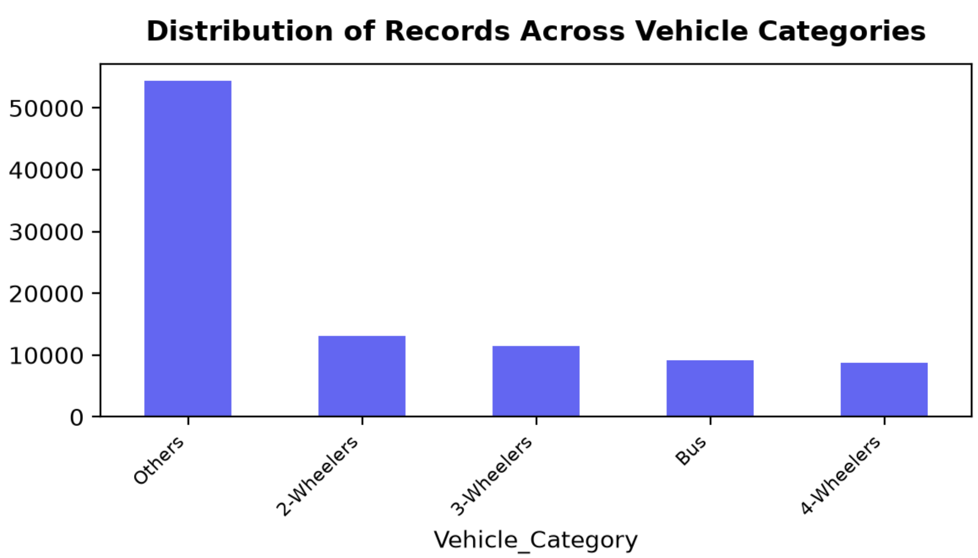
Column	Action	Explanation & Custom Rules
Date	DROP	Explicitly requested for removal by the user's goal.
EV_Sales_Quantity	DROP	Explicitly requested for removal by the user's goal to drop the 'quantity' column.
Month_Name	KEEP	Provides temporal context and is not explicitly requested for removal.
State	KEEP	Provides geographical context and is not requested for removal.
Vehicle_Category	KEEP	Provides categorical context about the vehicles and is not requested for removal.
Vehicle_Class	KEEP	Provides categorical context about the vehicles and is not requested for removal.
Vehicle_Type	KEEP	Provides categorical context about the vehicles and is not requested for removal.
Year	KEEP	Provides temporal context and is not explicitly requested for removal.

3. Cleaned Dataset Descriptive Statistics

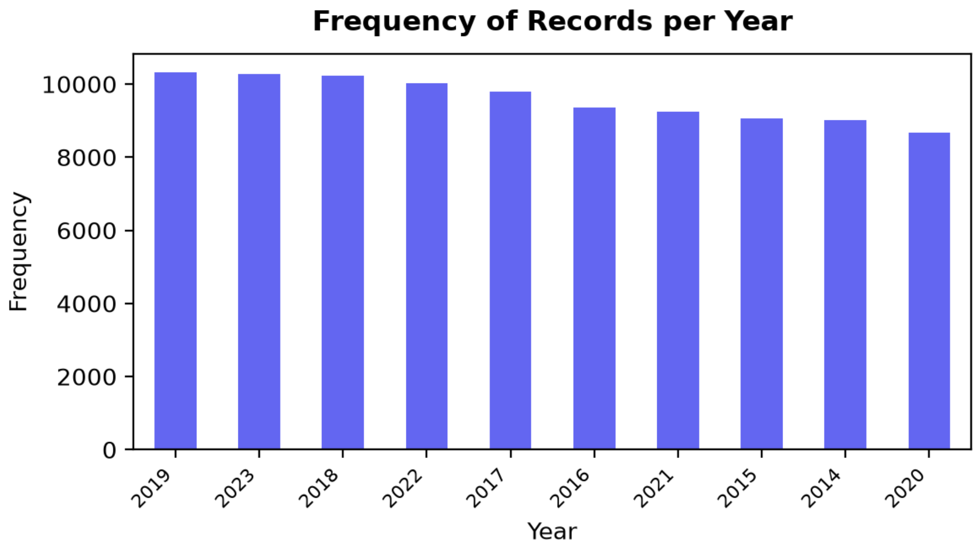
index	Year
count	96845.0
mean	2018.62
std	2.9
min	2014.0
25%	2016.0
50%	2019.0
75%	2021.0
max	2024.0

4. Diagnostic Visualizations

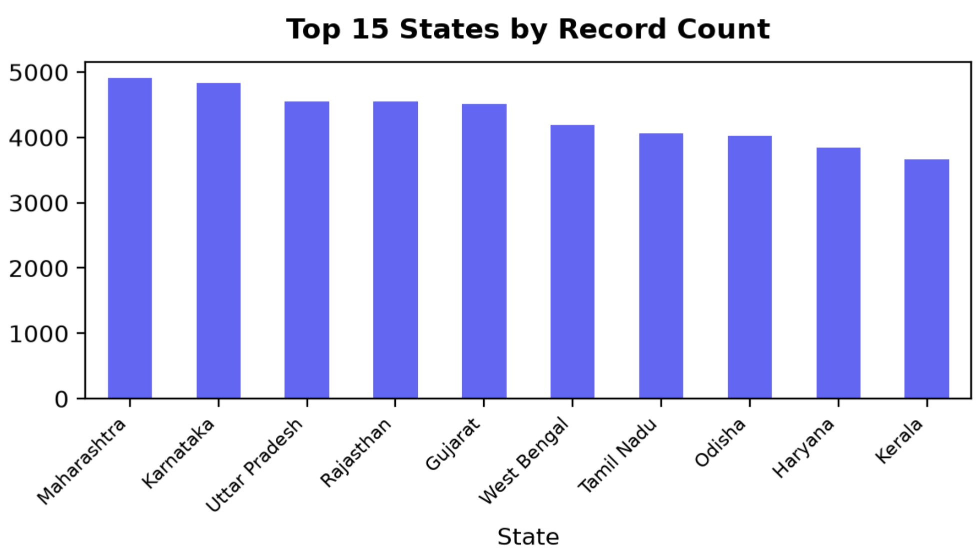
This bar chart aggregates the number of records for each of the 5 Vehicle Categories, providing a clear view of the most and least common vehicle groups in the cleaned dataset.



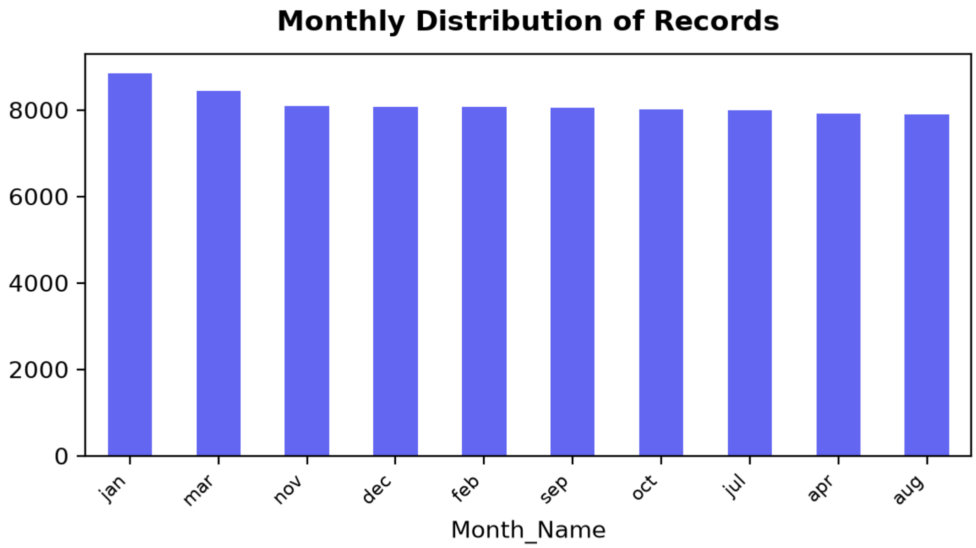
A histogram to visualize the distribution of records across the 11 available years, helping to identify trends or gaps in data collection over time.



This bar chart displays the count of records for the top 15 States out of the 34 unique values, highlighting geographical concentration and regional disparities in the dataset.



A bar chart showing the count of records for each of the 12 months, useful for identifying any seasonal patterns or uniformity in how the data was recorded throughout the year.



Monthly Distribution of Records
8000
6000
4000
2000
0
jan
mar
nov
dec
feb
sep
oct
jul
apr
aug
Month_Name